

Solution: Supplementary Problem 1

Chapter 10: Rotational Dynamics – Varied Motion

1. Preliminary Data & Conversions

To ensure accuracy, all physical quantities are converted into SI units:

- **Mass (M):** 2 kg
- **Radius (R):** $40\text{ cm} = 0.4\text{ m}$
- **Initial Angular Speed (ω_0):**

$$\omega_0 = 600\text{ turns/min} = 600 \times \frac{2\pi}{60} = \mathbf{20\pi\text{ rad/s}}$$

- **Angular Displacement ($\Delta\theta$):**

$$\Delta\theta = 2\text{ turns} = 2 \times 2\pi = \mathbf{4\pi\text{ rad}}$$

- **Moment of Inertia (I):** For a homogeneous disc rotating about its central axis:

$$I = \frac{1}{2}MR^2 = \frac{1}{2}(2)(0.4)^2 = \mathbf{0.16\text{ kg} \cdot \text{m}^2}$$

Question 1: Calculate the Braking Force f

A. Proof of Uniformly Varied Rotational Motion (U.V.R.M.)

Applying **Newton's Second Law for Rotation** about the axis (O):

$$\sum M_O(\vec{F}_{ext}) = I\alpha$$

The forces acting on the wheel are:

1. **Weight ($\vec{P}$):** $M_O(\vec{P}) = 0$ (passes through the axis).
2. **Reaction ($\vec{R}$):** $M_O(\vec{R}) = 0$ (passes through the axis).
3. **Braking Force ($\vec{f}$):** Tangential force acting at distance R opposing motion.

$$\sum M_O = M_O(\vec{f}) = -f \cdot R$$

Since the braking force f is constant and the radius R is constant, the resultant torque is constant. Furthermore, since the moment of inertia I is constant:

$$\alpha = \frac{\sum M_O}{I} = \frac{-f \cdot R}{I} = \text{constant}$$

The constant angular acceleration α proves that the wheel undergoes **Uniformly Varied Rotational Motion (U.V.R.M.)**.

B. Calculation of Angular Acceleration and Force

Using the **time-independent kinematic equation** for rotation:

$$\omega_f^2 - \omega_0^2 = 2\alpha\Delta\theta$$

Substituting $\omega_f = 0$ (the wheel stops):

$$0^2 - (20\pi)^2 = 2 \cdot \alpha \cdot (4\pi)$$

$$-400\pi^2 = 8\pi \cdot \alpha \implies \alpha = \frac{-400\pi^2}{8\pi} = -50\pi \text{ rad/s}^2$$

Substituting α back into the dynamics equation ($I\alpha = -fR$):

$$(0.16) \cdot (-50\pi) = -f \cdot (0.4)$$

$$-8\pi = -0.4f \implies f = \frac{8\pi}{0.4} = 20\pi \text{ N}$$

Braking Force: $f = 20\pi \approx 62.83 \text{ N}$

Question 2: How long does it take to stop?

To find the time t using angular displacement, we use the kinematic equation relating distance to average angular speed:

$$\Delta\theta = \frac{\omega_0 + \omega_f}{2} \cdot t$$

$$4\pi = \frac{20\pi + 0}{2} \cdot t$$

$$4\pi = 10\pi \cdot t \implies t = \frac{4\pi}{10\pi} = 0.4 \text{ s}$$

Stopping Time: $t = 0.4 \text{ s}$

Question 3: Magnitude of the Reaction Force R

Since the center of mass of the pulley remains stationary, the net translational force is zero ($\sum \vec{F} = 0$). In the vertical direction:

$$\vec{R} + \vec{P} + \vec{f} = 0$$

Given that the weight $\vec{P}$ and the force $\vec{f}$ are both directed downwards:

$$R = P + f = (M \cdot g) + f$$

$$R = (2 \cdot 10) + 20\pi = 20 + 62.83 = 82.83 \text{ N}$$

Reaction Force: $R \approx 82.83 \text{ N}$